

Question 1 — Password Correction

```
counter = 0

errors = 0

special = ['@', '!', '?', '%', '#', '$', '&']

for i in range(10):
    password = input('enter pass: ')
    counter += 1
    if len(password) < 8:
        print('the pass must be 8 char at least')
        errors += 1
        print('-----')
    if password[0] != password[0].upper():
        print('first letter must be Capitalize')
        errors += 1
        print('-----')
    if password[1:] != password[1:].lower():
        print('at least one lower word needed')
        errors += 1
        print('-----')
    if not any(j.isdigit() for j in password):
        print('at least one number required')
        errors += 1
        print('-----')
    if not any(j in '@!%$#' for j in password):
        print('at least one special chr needed (@ ! ? $ # & %)')
        errors += 1
```

```
print('-----')  
if errors == 0: print('valid password')
```

Question 2 — Remove Repeated Characters

```
words = input('type: ')  
result = ""  
for chrSlice in words:  
    if chrSlice not in result:  
        result += chrSlice  
print(result)
```

Question 3 — Letter Counter

```
text = input('Enter text: ')  
letters = 0  
uppercase = 0  
lowercase = 0  
digits = 0  
spaces = 0  
special = 0  
for char in text:  
    if char.isalpha():  
        letters += 1  
    if char.isupper():  
        uppercase += 1  
    else: lowercase += 1  
    elif char.isdigit():
```

```
digits += 1
elif char.isspace():
spaces += 1
else:
special += 1
print("Letters:", letters)
print("Uppercase:", uppercase)
print("Lowercase:", lowercase)
print("Digits:", digits)
print("Spaces:", spaces)
print("Special characters:", special)
```

Question 4 — Most Repeated Word

```
text = input()
words = text.split()
most_common = words[0]
max_count = words.count(words[0])
for word in words:
count = words.count(word)
if count > max_count:
max_count = count
most_common = word
print(most_common, "->", max_count)
```

Question 5 — Longest Word

```
text = input()
words = text.split()
longest = words[0]
for word in words:
    if len(word) > len(longest):
        longest = word
print(longest)
print("Length:", len(longest))
```

Question 6 — Find Specific Words

```
words = ["hack", "fraud", "scam", "password", "attack"]
text = input().lower()
for word in words:
    count = text.split().count(word)
    if count > 0:
        print(word, "->", count)
```

Question 7 — Character Compression

```
text = input()
result = ""
count = 1
for i in range(len(text)):
    if i + 1 < len(text) and text[i] == text[i + 1]:
        count += 1
    else:
        result += text[i] + str(count)
        count = 1
print(result)
```

Question 8 — Text Report

```
text = input()
words = text.split()

total_characters = len(text)
total_words = len(words)
total_letters = 0
total_digits = 0
total_spaces = 0
total_uppercase = 0
total_lowercase = 0
for char in text:
    if char.isalpha():
        total_letters += 1
    if char.isdigit():
        total_digits += 1
    if char == " ":
        total_spaces += 1
    if char.isupper():
        total_uppercase += 1
    if char.islower():
        total_lowercase += 1
longest_word = words[0]
shortest_word = words[0]
for word in words:
    if len(word) > len(longest_word):
        longest_word = word
```

```
if len(word) < len(shortest_word):
    shortest_word = word
most_repeated_character = max(set(text), key=text.count)
most_repeated_word = max(set(words), key=words.count)
print("Total characters:", total_characters)
print("Total words:", total_words)
print("Total letters:", total_letters)
print("Total digits:", total_digits)
print("Total spaces:", total_spaces)
print("Total uppercase:", total_uppercase)
print("Total lowercase:", total_lowercase)
print("Longest word:", longest_word)
print("Shortest word:", shortest_word)
print("Most repeated character:", most_repeated_character)
print("Most repeated word:", most_repeated_word)
```

Question 9 — Login System

```
correct_username = "admin"
```

```
correct_password = "1234"
```

```
attempts = 3
```

```
while attempts > 0:
```

```
    username = input("Username: ")
```

```
    password = input("Password: ")
```

```
    if username == correct_username and password == correct_password:
```

```
        print("Login successful")
```

```
        break
```

```
    else:
```

```
        attempts -= 1
```

```
        print("Wrong user name or password.")
```

```
        print("Attempts remaining:", attempts)
```


Question 10 — Common Words

```
sentence1 = input()
sentence2 = input()
words1 = sentence1.split()
words2 = sentence2.split()
common_words = []
for word in words1:
    if word in words2 and word not in common_words:
        common_words.append(word)
print("Common words:")
for word in common_words:
    print(word)
```